Patch Positivity and Comparison Principles

A transcription of the handwritten presentation

Editorial convention. The order and content of the six source pages are preserved. Clearly crossed-out drafts are omitted, and ordinary spelling and unambiguous indexing slips are regularized. Blue markers refer to transcription notes at the end; these identify unresolved readings and mathematical discrepancies. The attached paper is used only to clarify notation and record discrepancies.

1. Interacting particle systems and ergodicity

Handwritten page 1

- Configurations: $\eta \in \{0,1\}^{\Lambda}$, with Λ at most countable.
- $\eta \mapsto \eta^i$: the configuration obtained by flipping site i.
- Generator:
$$\mathcal{L}f(\eta) = \sum_{i\in\Lambda} c_i(\eta)\big(f(\eta^i) - f(\eta)\big), \qquad c_i(\eta) \text{ is the flip rate.}$$
- Example: independent Bernoulli flips,
$$\mathcal{L}f(\eta) = \sum_{i\in\Lambda} \big(\mathbb{E}_{i,p_i(\eta)}[f](\eta) - f(\eta)\big).$$
Here $\mathbb{E}_{i,r}$ means refresh site i to Bern(r). For independent flips the refresh parameters do not depend on η.
- η_0: initial configuration. η_t: evolution.

Central question. Is the IPS ergodic? Is there a unique stationary measure π such that, for every initial configuration η_0,
$$\eta_t \Rightarrow \pi, \qquad |\mathbb{E}[f(\eta_t) \mid \eta_0] - \mathbb{E}_\pi[f]| \longrightarrow 0 \quad \text{for every local } f?$$

Tool: monotonicity. An IPS is monotone if ordered initial configurations admit an order-preserving coupling:
$$\eta_0 \le \xi_0 \quad \Longrightarrow \quad \exists \text{ a coupling with } \eta_t \le \xi_t \quad (t \ge 0).$$

Examples: independent Bernoulli flips, ferromagnetic Ising dynamics, and the contact process.

Comparison principle. In such a coupling,
$$\eta_t\,\big|_{\eta_0=\mathbf{0}} \le\ \eta_t\,\big|_{\eta_0} \le\ \eta_t\,\big|_{\eta_0=\mathbf{1}}\,.$$

The extremal evolutions converge to π^- and π^+, respectively. Hence, if $\pi^- = \pi^+$, the process is ergodic.

Question. Ergodicity for nonmonotone systems? *[The short trailing handwritten parenthesis is unclear.]*

2. Breaking monotonicity: kinetically constrained spin models

Handwritten page 2

$$\mathcal{L}f(\eta) = \sum_i \lambda_i(\eta)\big(\mathbb{E}_{i,p_i(\eta)}[f](\eta) - f(\eta)\big), \qquad \lambda_i(\eta) \in \{0,1\}.$$

Here λ_i is the constraint indicator.

Example: FA-1f (written "FA-1n" on some source pages).

$$\lambda_i(\eta) = \begin{cases} 1, & \text{at least one neighbour is 1,} \\ 0, & \text{otherwise.} \end{cases}$$ [T1]

A toy model for freezing and melting. The Bernoulli product measure π_p is invariant. The neighbourhood sketches show a central site surrounded by zeros ("frozen") and a configuration with a neighbouring 1 ("unfrozen").[T1]

East model. $\Lambda = \mathbb{Z}$. Bernoulli-p update if the right neighbour is 1; otherwise frozen.[T1]

Not monotone. The source illustrates the failure by

$$\begin{array}{ccccc} 00 & \leq & 01 & \text{(frozen)} \\ \downarrow & & \| & \\ 10 & \not\leq & 01. & \end{array}$$

This sketch uses 0 as the facilitating state.[T1]

Conjecture 1 (folklore). FA-1f is "ergodic" whenever

$$\exists i : \eta_0(i) = 0, \qquad \text{i.e. } \eta_0 \neq \mathbf{0} \text{ (for every } i\text{).}$$ [T2]

The margin distinguishes "on $\mathbb{Z}$" and "on $\mathbb{Z}^d$". A nearby reference reads "[Martinelli?], Shapira, Toninelli, '25".[T3]

Known result. Theorem 1 [Cancrini, Martinelli, Roberto, Toninelli, 2008]: FA-1f has a spectral gap for any $p < 1$.[T3]

- Not good enough for infinite volume.
- No (uniform) log-Sobolev inequality.
- Blondel et al., 2013.
- "High facilitating regime" (analogy: Ising at high temperature).

Conjecture 2. East model plus uniform deaths is ergodic.

- A missing case of the positive-rates conjecture for simple IPS.
- A final margin note mentions "2–3 years"; the preceding word is unclear.

3. Noise, ergodicity, and the monomial representation

Handwritten page 3

Question. Does adding noise cause, facilitate, or preserve ergodicity? A name in the margin appears to read "Jonas Köppl".[T3]

Idea. In a KCSM, adding 0s (vacancies) unblocks frozen configurations.

Difficulty.

- FA-1f without noise: π is an independent Bernoulli law $\Longrightarrow$ spectral-gap methods.
- FA-1f plus noise: no explicit formula for $\pi \Longrightarrow$ "no spectral gap".[T4]

A comparison principle for nonmonotone spin systems; patch positivity; FA-1f plus noise; a "coupling method".

Main result: Theorem (Głuchowski and Menz, 2026).[T5] Given

- η_t: FA-1f plus $\varepsilon\,\mathrm{Bern}(s)$ noise,
- $s \geq p$ ("we would like to get rid of" this restriction),
- $\inf_i(1-s_i) > 0$ (uniform rate of deaths),

then η_t is ergodic if $p \leq \frac{1}{2}$ or the interactions are orientable (for example, the East model).

Margin notes: $s \geq p$ corresponds to "melting" and $s \leq p$ to "freezing"; $p \leq \frac{1}{2}$ is the "high facilitating regime".[T6]

Patch-positive spin systems

Vector space. Local functions $f : \{0,1\}^\Lambda \to \mathbb{R}$.

Basis. For finite $S \subseteq \Lambda$,

$$\chi_S(\eta) = \prod_{i\in S} \eta(i) = \begin{cases} 1, & S \text{ is occupied,} \\ 0, & \text{otherwise.} \end{cases}$$

Every local function has the expansion

$$f(\eta) = \sum_{S\Subset\Lambda} \widehat{f}(S)\chi_S(\eta), \qquad \widehat{f}(S) = \sum_{B\subseteq S} (-1)^{|S\setminus B|} f(\mathbf{1}_B),$$

where $\mathbf{1}_B$ is 1 on B and 0 elsewhere.

Rewrite the generator.

$$\mathcal{L}f(\eta) = \sum_{i\in\Lambda} c_i(\eta)\big(f(\eta^i) - f(\eta)\big) = \sum_{i\in\Lambda} \lambda_i(\eta)\big(\mathbb{E}_{i,p_i(\eta)}[f](\eta) - f(\eta)\big).$$

Why two parameters? At each site there are two rates, $\eta(i) : 0 \to 1$ and $\eta(i) : 1 \to 0$.

$$\lambda_i(\eta) = \sum_{S\Subset\Lambda} \widehat{\lambda}_i(S)\chi_S(\eta), \qquad p_i(\eta) : \text{ Bernoulli refresh parameter.}$$

4. Patch positivity, the threshold profile, and examples

Handwritten page 4

Definition (patch positivity). An IPS is patch-positive if, for every i and every $\varnothing \neq S \subseteq N(i)$,

$$\widehat{\lambda}_i(S) \leq 0, \qquad \lambda_i\big(\widehat{p_i - p_i}(\mathbf{0})\big)(S) \geq 0.$$

Here $N(i)$ denotes the neighbours of i, $p_i = p_i(\eta)$ is a function of η, and $p_i(\mathbf{0})$ is its value at the all-zero configuration.

Definition (threshold profile).

$$p_i^\star := \max\left(\{0\} \cup \left\{\frac{\widehat{\lambda_i p_i}(S)}{\widehat{\lambda}_i(S)} : \varnothing \neq S \subseteq N(i),\ \widehat{\lambda}_i(S) < 0\right\}\right).$$

Distinguishes melting and freezing regimes. If $p_i(\eta) = p_i$ is constant, the notes state $p_i^\star = p_i$.[T7]

Examples.

1. **Independent Bernoulli updates:**

$$\mathcal{L}f = \sum_i \big(\mathbb{E}_{i,p_i}[f] - f\big), \qquad \lambda_i = 1,$$

$$\widehat{\lambda}_i(\varnothing) = 1, \qquad \widehat{\lambda}_i(S) = 0 \quad (S \neq \varnothing),$$

$$\lambda_i p_i = p_i, \qquad \widehat{\lambda_i p_i}(\varnothing) = p_i, \qquad \widehat{\lambda_i p_i}(S) = 0 \quad (S \neq \varnothing).$$

   The handwritten conclusion is $p_i^\star = p_i$.[T7]
2. **FA-1f:**

$$\lambda_i = 1 - \chi_{N(i)}, \qquad p_i = p_i(\mathbf{0}) \quad \text{(independent of } \eta).$$

   Hence patch-positive.
3. **FA-nf, $n \geq 1$:** not patch-positive.[T8]
4. **Biased annihilating branching process:**

$$\lambda_i(\eta) = \sum_{j \in N(i)} (1 - \eta(j)), \qquad \widehat{\lambda}_i(\{j\}) = -1 \quad (j \in N(i)),$$

   with $p_i(\eta) = p_i$ constant. Patch-positive.
5. **Contact process:**

$$\lambda_i(\eta) = 1 + \beta \sum_{j \in N(i)} \eta(j).$$

   Its nonconstant coefficients are positive. Relabel 0 and 1 to obtain a patch-positive system.

Inset: perturbed FA-1f.

$$\mathcal{L}^{\varepsilon,\boldsymbol{s}} f(\eta) = \sum_i \left\{\lambda_i(\eta)\big(\mathbb{E}_{i,p_i}[f](\eta) - f(\eta)\big) + \varepsilon\big(\mathbb{E}_{i,s_i}[f](\eta) - f(\eta)\big)\right\},$$

$$\lambda_i^{\varepsilon,\boldsymbol{s}} = \lambda_i + \varepsilon, \qquad p_i^{\varepsilon,\boldsymbol{s}} = \frac{p_i\lambda_i + \varepsilon s_i}{\lambda_i + \varepsilon}.$$

5. Perturbations and the comparison principle

Handwritten page 5

Remark. Patch positivity $\neq$ monotonicity.

Perturbed FA-1f. For constant p and s,

$$\mathcal{L}^{\varepsilon,s} f = \sum_i \{\lambda_i(\mathbb{E}_{i,p} f - f) + \varepsilon(\mathbb{E}_{i,s} f - f)\}, \quad \lambda_i^{\varepsilon,s} = \lambda_i + \varepsilon, \quad p_i^{\varepsilon,s} = \frac{p\lambda_i + \varepsilon s}{\lambda_i + \varepsilon}.$$

For $S \neq \varnothing$,

$$\widehat{\lambda_i^{\varepsilon,s}}(S) = \widehat{\lambda}_i(S), \qquad \widehat{\lambda_i^{\varepsilon,s} p_i^{\varepsilon,s}}(S) = p\widehat{\lambda}_i(S).$$

Hence

$$\lambda_i^{\varepsilon,s}\big(\widehat{p_i^{\varepsilon,s} - p_i^{\varepsilon,s}}(\mathbf{0})\big)(S) = \big(p - p_i^{\varepsilon,s}(\mathbf{0})\big)\widehat{\lambda}_i(S).$$

The second factor is nonpositive, and the first is nonpositive when $s \geq p$.

Lemma. Patch-positive $\iff s \geq p$; then $p^\star = p$.

Observation. Removing deaths preserves patch positivity ("wrong direction for applications").

Comparison principle

Definition (centered monomial).

$$\chi_S^\star(\eta) = \prod_{i \in S} (\eta(i) - p_i^\star).$$

Centered monomial order (CMO).

$$\mu \preceq_\star \nu \quad \iff \quad \mathbb{E}_\mu[\chi_A^\star] \leq \mathbb{E}_\nu[\chi_A^\star] \quad \text{for every finite } A \subseteq \Lambda.$$

Lemma. If $\mu \preceq_\star \nu$, then $\mathbb{E}_\mu[\chi_S] \leq \mathbb{E}_\nu[\chi_S]$ for every finite S.

Proof. Expanding the product gives

$$\begin{aligned}\chi_S(\eta) &= \prod_{i \in S} \big((\eta(i) - p_i^\star) + p_i^\star\big) \\ &= \sum_{A \subseteq S} \prod_{i \in A} (\eta(i) - p_i^\star) \prod_{j \in S \setminus A} p_j^\star = \sum_{A \subseteq S} \chi_A^\star(\eta) \prod_{j \in S \setminus A} p_j^\star.\end{aligned}$$

Since each coefficient is nonnegative,

$$\mathbb{E}_\mu[\chi_S] = \sum_{A \subseteq S} \mathbb{E}_\mu[\chi_A^\star] \prod_{j \in S \setminus A} p_j^\star \leq \sum_{A \subseteq S} \mathbb{E}_\nu[\chi_A^\star] \prod_{j \in S \setminus A} p_j^\star = \mathbb{E}_\nu[\chi_S]. \qquad \square$$

Theorem (comparison principle). Patch-positive IPS preserve CMO:

$$\mu \preceq_\star \nu \quad \Longrightarrow \quad \mu P_t \preceq_\star \nu P_t.$$

Allows sandwiching. *[A further "And if" line is crossed out and incomplete.]*

6. Ergodicity and the proof mechanism

Handwritten page 6

Theorem (Głuchowski and Menz, 2026).[T5] Given

- a patch-positive process η_t,
- uniform pure deaths at rate at least $\varepsilon > 0$,
- the class

$$\mathcal{M}_- = \left\{\mu : \exists K < \infty \; \forall A \Subset \Lambda, \quad \mu(\chi_A^\star) \geq -K(1-p^\star)^{|A|}\right\},$$

then η_t is ergodic on $\mathcal{M}_-$. The displayed definition uses a constant threshold $p^\star$.[T9]

Observation. If $p^\star \leq \frac{1}{2}$, then $\mathcal{M}_- = \mathcal{P}(\Omega)$, where $\Omega = \{0,1\}^\Lambda$. The handwritten justification is

$$\mu(\chi_A^\star) \geq -(p^\star)^{|A|} \geq -(1-p^\star)^{|A|}.$$[T10]

Corollary. *[The source leaves this as "..."]*

Question. Ergodicity of FA-1f, or orientable systems?[T11]

- Burn-in period: the evolving law gets close to $\mathcal{M}_-$.

Remark. For patch-positive spin systems, noise causes ergodicity.[T5]

Technical mechanism

The handwritten proof outline links the following steps:

1. Patch positivity and a dual signed set-valued process.
2. A Feynman–Kac formula.
3. Independence after conditioning on a system of clocks.
4. Patches and products of patch contributions.
5. Nonnegative patch contributions.
6. Comparison.

Message. For patch-positive systems, adding vacancies causes ergodicity in the "high density regime".[T12]

Transcription notes

These notes are editorial and are not part of the handwritten talk. They separate the source's wording from corrections or qualifications supported by the attached paper. No new literature verification has been performed.

T1. Facilitating-spin convention (page 2). The handwritten FA-1f and East definitions say that a neighbouring 1 facilitates, and the neighbourhood sketches agree with that wording. However, the nonmonotonicity example, the later formula $\lambda_i = 1 - \chi_{N(i)}$, and the attached paper all use a neighbouring 0 as facilitating. With the paper's convention, replace "is 1" by "is 0" in both definitions; the frozen/unfrozen sketch labels must also be adjusted. The transcription retains the source's contradiction explicitly.

T2. Initial state in Conjecture 1 (page 2). The source has $\exists i : \eta_0(i) = 0$ but its parenthetical appears to say $\eta_0 \neq \mathbf{0}$. With 0 facilitating, the equivalent condition is $\eta_0 \neq \mathbf{1}$. The phrase "for every i" in the parenthesis is also inconsistent with the existential condition. "Ergodic" is quoted in the source; here the intended question is convergence from the specified initial states, not from the absorbing all-one state.

T3. Names and abbreviated references (pages 2–3). The first surname beside "Shapira, Toninelli, '25" is not fully legible; "Martinelli" is a tentative reading. The margin name on page 3 is also tentative. The other names and years are transcribed as written, without adding bibliographic details or verifying the literature claims.

T4. "No spectral gap" (page 3). This is the source's shorthand. A lack of an explicit invariant measure does not imply that a spectral gap is absent. It makes the usual spectral-gap approach harder to apply. No nonexistence claim is justified by that line alone.

T5. Scope of the ergodicity statements (pages 3 and 6). The talk abbreviates hypotheses. In the attached paper the general convergence result assumes polynomial growth of the neighbourhood graph, patch positivity, and a uniformly positive pure-death component, within the paper's setup of bounded local rates. The threshold bound $\boldsymbol{p}^\star \leq \frac{1}{2}$ gives uniform exponential ergodicity; orientability gives uniform ergodicity. The headline "noise causes ergodicity" is to be read under these hypotheses. For Bernoulli noise of rate ε, the added pure-death rate is $\varepsilon(1 - s_i)$. Page 6 instead uses ε directly for a lower bound on the pure-death rate.

T6. Melting/freezing labels (page 3). With 0 vacant and a Bernoulli parameter equal to the probability of 1, $s \geq p$ biases the added refresh toward occupation rather than vacancies. The handwritten "melting" and "freezing" labels therefore appear reversed relative to the convention used in the paper.

T7. Threshold for independent refreshes (page 4). Under the displayed definition, independent refreshes have $p_i^\star = \max\{0\} = 0$, because all nonconstant coefficients of λ_i vanish. The handwritten $p_i^\star = p_i$ is therefore not the minimal threshold specified by that definition. For constant refresh parameter p_i, the equality $p_i^\star = p_i$ follows when at least one nonconstant coefficient of λ_i is strictly negative, as in FA-1f with a nonempty neighbourhood.

T8. FA-nf (page 4). The handwritten range appears to be $n \geq 1$. The attached paper states $n \geq 2$ (with the usual nontrivial constraint), which avoids contradicting the preceding FA-1f example.

Transcription notes (continued)

T9. The class $\mathcal{M}_-$ (page 6). The first symbol in the handwritten set is unclear; μ is used because this is a class of probability laws. For a nonconstant threshold profile, the corresponding formulation in the paper is

$$\mathcal{M}_- = \left\{ \mu : \exists K \geq 0 \ \forall A \Subset \Lambda, \quad \mu(\chi^\star_A) \geq -K \prod_{i \in A} (1 - p^\star_i) \right\}.$$

The abbreviation "ergodic on $\mathcal{M}_-$" means that every initial law in this class converges to a common invariant measure.

T10. The lower bound on centered moments (page 6). The first inequality in the handwritten justification is generally false. For example, when $p^\star = \frac{1}{4}$ and A contains one zero spin and one one spin, $\chi^\star_A = -3/16 < -1/16 = -(p^\star)^2$. The paper instead uses the valid pointwise estimate

$$\chi^\star_A(\eta) \geq -\prod_{i \in A} (1 - p^\star_i) \qquad \text{when } p^\star_i \leq \tfrac{1}{2} \text{ for all } i.$$

It follows from $|\eta(i) - p^\star_i| \leq 1 - p^\star_i$. Integrating gives $\mu \in \mathcal{M}_{-,1}$ for every probability measure μ, and hence the stated conclusion $\mathcal{M}_- = \mathcal{P}(\Omega)$.

T11. Orientability and burn-in (page 6). The short question after "FA-1f" is not fully legible; "or orientable" is the most likely reading. The source appears to say that burn-in approaches $\mathcal{M}_-$. In the attached paper's orientability proof, the class approached is $\mathcal{M}_\star$, where

$$\mathcal{M}_\star = \{\mu : \mu(\chi^\star_A) \geq 0 \text{ for every finite } A\}.$$

The transcription preserves the apparent handwritten subscript.

T12. Closing phrase (page 6). "High density regime" is legible but does not specify the density of which spin. Earlier the talk calls $p^\star \leq \frac{1}{2}$ the "high facilitating regime". With 0 facilitating, this corresponds to high vacancy density, not high occupation density. The closing sentence is retained as written.